



Kimia Farma Business Performance 2020 – 2023

Kimia Farma – Big Data Analytics

Presented By: Debby Rofiko Malik



Debby Rofiko Malik

Data Enthusiast



A highly adaptable Biomedical Engineering graduate from Universitas Indonesia with strong analytical skills and solid problem-solving abilities. Experienced in turning raw data into clear, actionable insights using Excel, SQL, and Looker Studio. A fast learner with a firm belief in the value of data for informed decision-making and sustainable growth, committed to continuous learning and delivering effective data-driven solutions.



West Jakarta



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Courses & Certifications



**Data Analytics & Visualization
Bootcamp**

| [Link Certificate](#)

(November, 2025)



Data Science & Data Analysis

| [Link Certificate](#)

(October, 2025)



Machine Learning with Python

| [Link Certificate](#)

(June, 2024)

About the Company

Kimia Farma Tbk is **Indonesia's first pharmaceutical company**, founded in 1817 and now part of the state-owned enterprise (BUMN) group. The company operates in **pharmaceutical manufacturing, distribution, and retail services**. With an extensive network of pharmacies and branches across the country, Kimia Farma **plays a significant role in providing the public with accessible, high-quality, and affordable healthcare products**.



Final Project Summary

The final assignment of the internship required an **in-depth examination of actual transactional data from 2020 to 2023**. This thorough analysis aimed **to uncover essential insights that would aid in making more informed and strategic business choices**. The focus was on evaluating revenue trends over the years, identifying the best-performing branches, highlighting discrepancies between branch performance and customer ratings, and analyzing the distribution of profits across different provinces.



Problem Statement

How can Kimia Farma leverage transactional data from 2020–2023 to evaluate branch performance, understand profit distribution across regions, and uncover product and customer insights that inform more effective business strategies?



Outcome Highlight

The final dashboard presents a comprehensive view of business performance by highlighting top-performing branches, visualizing profit distribution across provinces, and uncovering key product and customer insights. These findings enable Kimia Farma to identify growth opportunities, strengthen product strategies, and better understand customer behavior to support data-driven decision-making.



Repo & Walk-through



[Github](#)

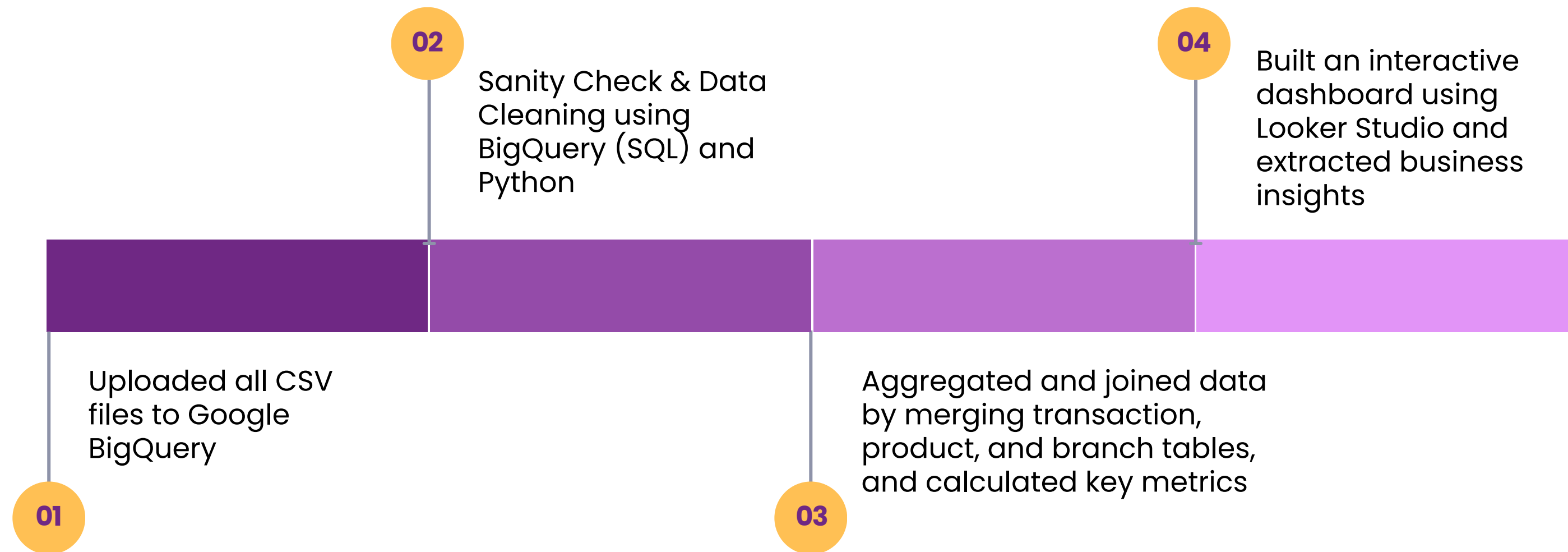


[Youtube](#)



[Looker Studio](#)

Project Workflow



Dataset & Entity Relationship Diagram



kf_product: provides product names, categories, and corresponding base prices



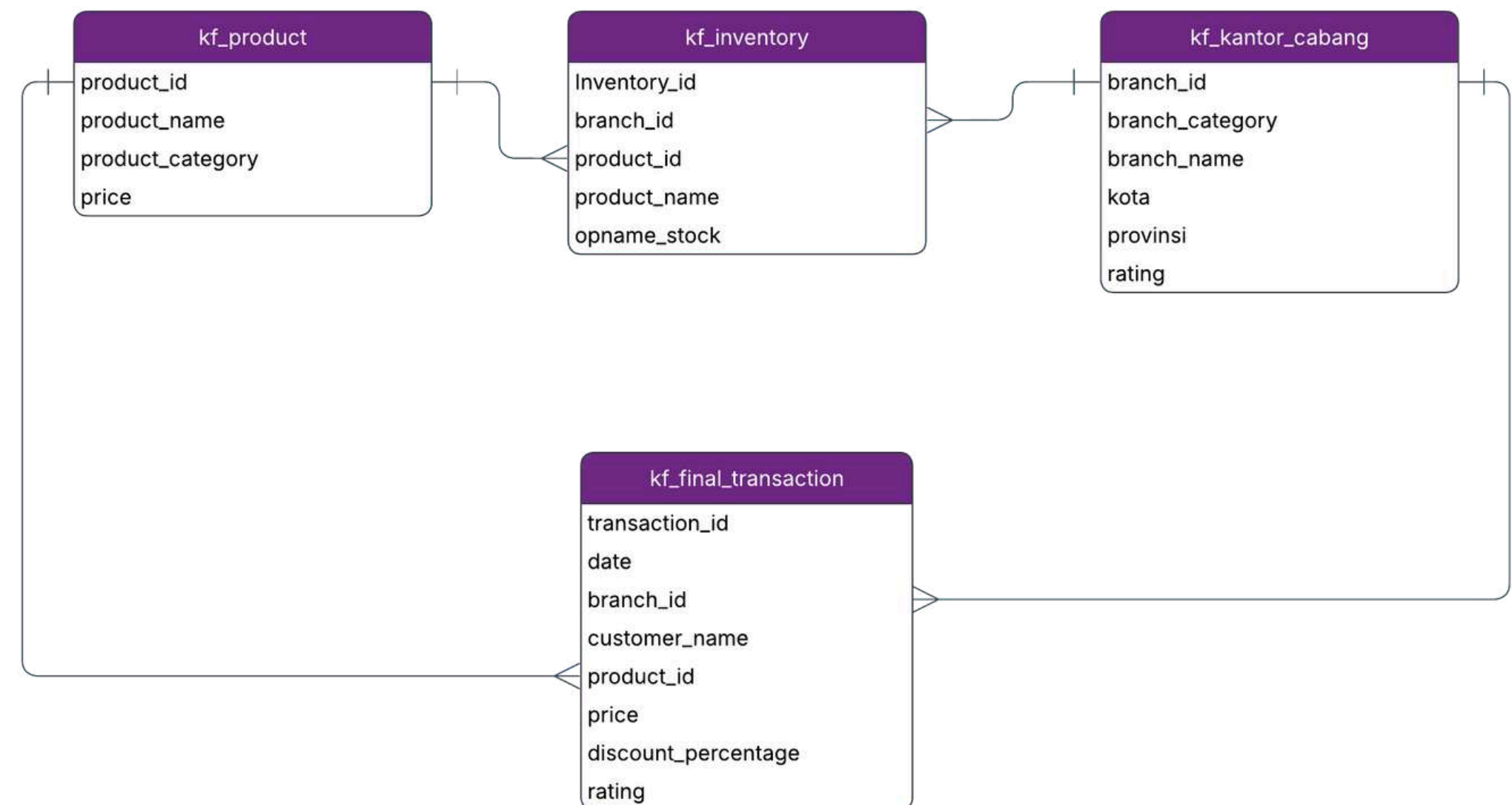
kf_kantor_cabang: branch-level data including location and ratings



kf_inventory: includes stock levels per branch (excluded from analysis as it was not relevant)



kf_final_transaction: contains transactions records including price, discount, and customer ratings



Data Sanity Check

Check for Duplicates: **NO DUPLICATES**

Tools: BigQuery.

```
1  -- Check Duplicate
2  -- 1. kf_final_transaction
3
4  SELECT
5      transaction_id, product_id, branch_id, customer_name,
6      date, price, discount_percentage, rating,
7      COUNT(*) AS CNT
8  FROM `rakamin-kf-analytics-479117.kimia_farma.kf_final_transaction`
9  GROUP BY
10     transaction_id, product_id, branch_id, customer_name,
11     date, price, discount_percentage, rating
12  HAVING COUNT(*) > 1;
13
14  -- 2. kf_product
15
16  SELECT
17      product_id, product_name, product_category, price,
18      COUNT(*) AS CNT
19  FROM `rakamin-kf-analytics-479117.kimia_farma.kf_product`
20  GROUP BY
21     product_id, product_name, product_category, price
22  HAVING COUNT(*) > 1;
23
24  -- 3. kf_inventory
25  SELECT
26      Inventory_ID, branch_id, product_id, product_name, opname_stock,
27      COUNT(*) AS CNT
28  FROM `rakamin-kf-analytics-479117.kimia_farma.kf_inventory`
29  GROUP BY
30     Inventory_ID, branch_id, product_id, product_name, opname_stock
31  HAVING COUNT(*) > 1;
32
33  -- 4. kf_cantor_cabang
34  SELECT
35      branch_id, branch_category, branch_name,
36      kota, provinsi, rating,
37      COUNT(*) AS CNT
38  FROM `rakamin-kf-analytics-479117.kimia_farma.kf_kantor_cabang`
39  GROUP BY
40     branch_id, branch_category, branch_name,
41     kota, provinsi, rating
42  HAVING COUNT(*) > 1;
```


Data Sanity Check

Check for Missing Values: **NO NULL**

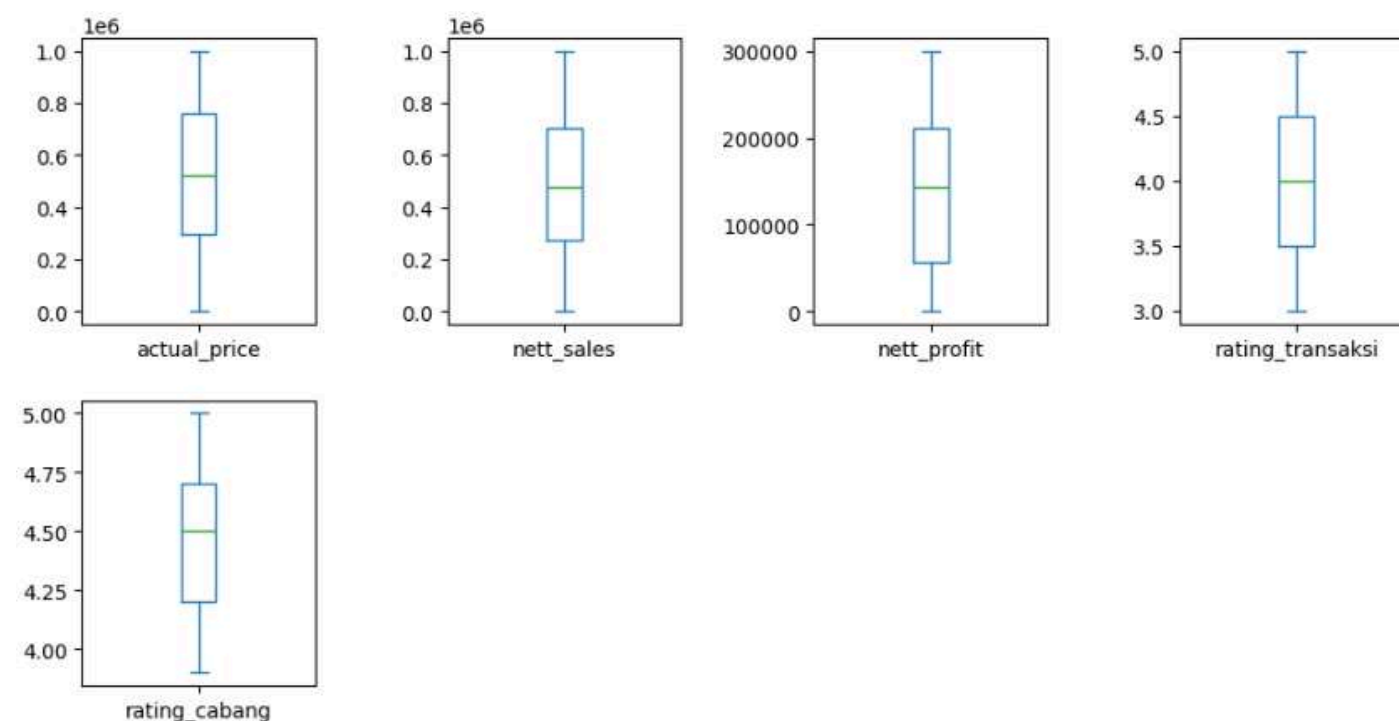
Tools: BigQuery.

```
47 -- Check Missing Values
48 -- 1. kf_final_transaction
49
50 SELECT *
51 FROM `rakamin-kf-analytics-479117.kimia_farma.kf_final_transaction` AS src
52 WHERE TO_JSON_STRING(src) LIKE '%null%';
53
54 -- 2. kf_product
55 SELECT *
56 FROM `rakamin-kf-analytics-479117.kimia_farma.kf_product` AS src
57 WHERE TO_JSON_STRING(src) LIKE '%null%';
58
59 -- 3. kf_inventory
60 SELECT *
61 FROM `rakamin-kf-analytics-479117.kimia_farma.kf_inventory` AS src
62 WHERE TO_JSON_STRING(src) LIKE '%null%';
63
64 -- 4. kf_kantor cabang
65 SELECT *
66 FROM `rakamin-kf-analytics-479117.kimia_farma.kf_kantor_cabang` AS src
67 WHERE TO_JSON_STRING(src) LIKE '%null%';
```


Data Sanity Check

Check for Outliers: **NO OUTLIERS**

Tools: Python




```
CREATE TABLE `rakamin-kf-analytics-479117.kimia_farma.kf_tabel_analisa` AS

WITH MAIN AS (
  SELECT
    ft.transaction_id,
    ft.date,
    ft.branch_id,
    cb.branch_name,
    cb.kota,
    cb.provinsi,
    cb.rating AS rating_cabang,
    ft.customer_name,
    pd.product_id,
    pd.product_name,
    ft.price AS actual_price,
    ft.discount_percentage,

    -- Persentase Gross Laba
    CASE
      WHEN ft.price <= 50000 THEN 0.10
      WHEN ft.price <= 100000 THEN 0.15
      WHEN ft.price <= 300000 THEN 0.20
      WHEN ft.price <= 500000 THEN 0.25
      ELSE 0.30
    END AS persentase_gross_laba,

    -- Nett Sales
    ft.price * (1 - ft.discount_percentage) AS nett_sales,

    -- Net Profit
    (ft.price * (1 - ft.discount_percentage)) *
    CASE
      WHEN ft.price <= 50000 THEN 0.10
      WHEN ft.price <= 100000 THEN 0.15
      WHEN ft.price <= 300000 THEN 0.20
      WHEN ft.price <= 500000 THEN 0.25
      ELSE 0.30
    END AS nett_profit,

    ft.rating AS rating_transaksi
  FROM `rakamin-kf-analytics-479117.kimia_farma.kf_final_transaction` AS ft
  LEFT JOIN `rakamin-kf-analytics-479117.kimia_farma.kf_kantor_cabang` AS cb
    ON ft.branch_id = cb.branch_id
  LEFT JOIN `rakamin-kf-analytics-479117.kimia_farma.kf_product` AS pd
    ON ft.product_id = pd.product_id
)
```

Tabel Analisa

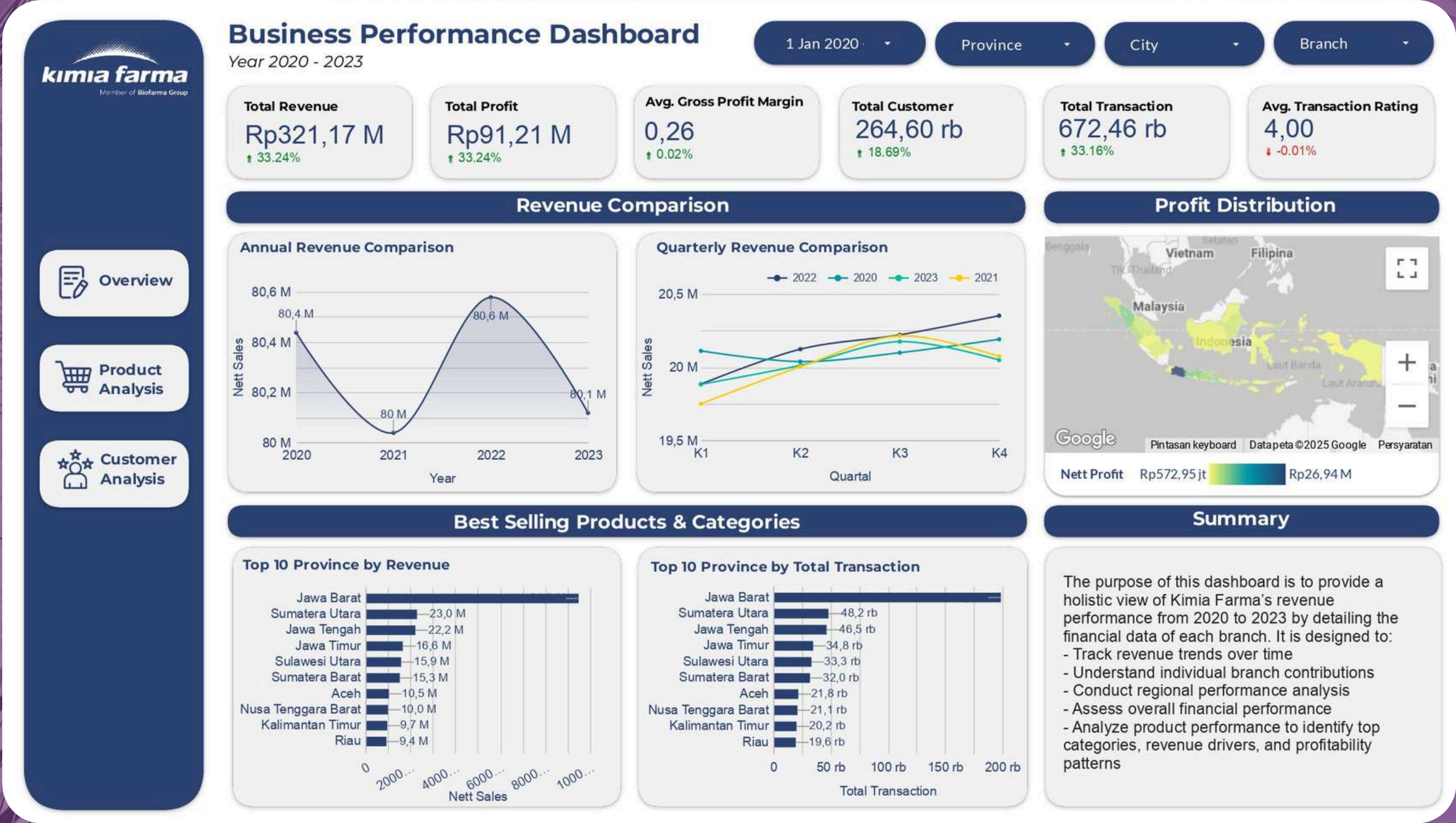


☐	Field name	Type	Mode
☐	transaction_id	STRING	NULLABLE
☐	date	DATE	NULLABLE
☐	branch_id	INTEGER	NULLABLE
☐	branch_name	STRING	NULLABLE
☐	kota	STRING	NULLABLE
☐	provinsi	STRING	NULLABLE
☐	rating_cabang	FLOAT	NULLABLE
☐	customer_name	STRING	NULLABLE
☐	product_id	STRING	NULLABLE
☐	product_name	STRING	NULLABLE
☐	actual_price	INTEGER	NULLABLE
☐	discount_percentage	FLOAT	NULLABLE
☐	persentase_gross_laba	FLOAT	NULLABLE
☐	nett_sales	FLOAT	NULLABLE
☐	nett_profit	FLOAT	NULLABLE
☐	rating_transaksi	FLOAT	NULLABLE

Data Aggregation and Joining

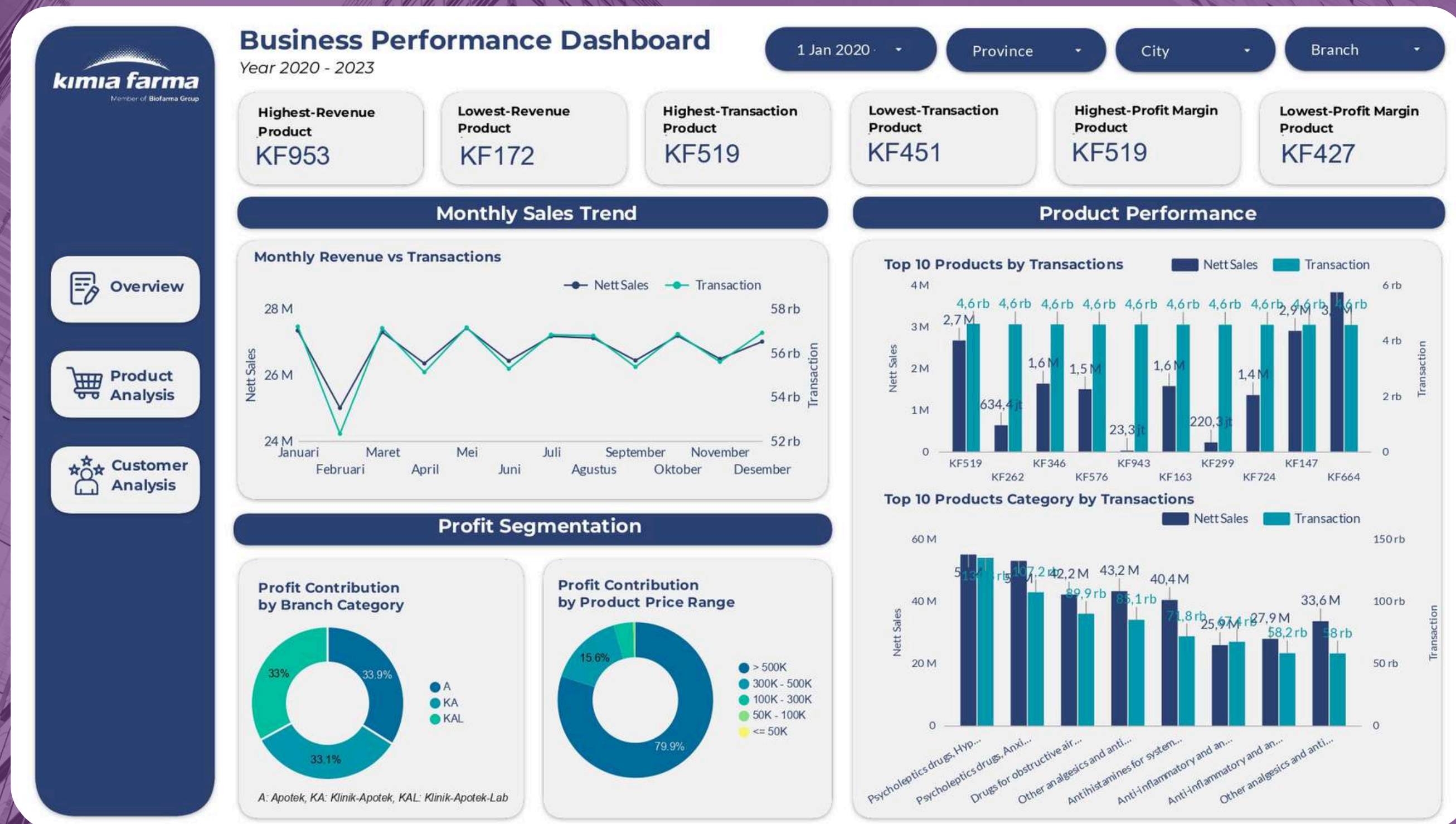
The gross profit percentage was calculated according to the price-based guidelines provided in the brief. This step creates a new table (kf_table_analisa) in BigQuery after performing aggregation and joins.

Dashboard Performance Analytics
Overview Page (1/3).



Dashboard Performance Analytics

Product Analysis Page (2/3)





Overview

Product Analysis

Customer Analysis

Business Performance Dashboard

Year 2020 - 2023

1 Jan 2020

Province

City

Branch

Average Basket Size	Average Basket Value	Average Profit per Transaction	Customer Lifetime Value
1,0	Rp477,61 rb	Rp135,64 rb	Rp91,21 M
	↑ 0.1%	↑ 0.1%	↑ 33.2%

Customer Segmentation

Customer Segmentation Based on Transactions

Customer - ...	Tr...	Avg...	Segmentation	Disc (A...	Trans. R...
1. Michael Smith - ...	13	Rp604.579,38	Loyal Customer	5,54%	3,99
2. John Smith - B...	12	Rp506.015,00	Loyal Customer	9,75%	3,97
3. Robert Johnson...	11	Rp472.602,00	Loyal Customer	7,82%	4,37
4. Michael Smith - ...	11	Rp453.988,27	Loyal Customer	5,55%	4,21
5. James Smith - ...	11	Rp420.428,55	Loyal Customer	8,18%	4,18

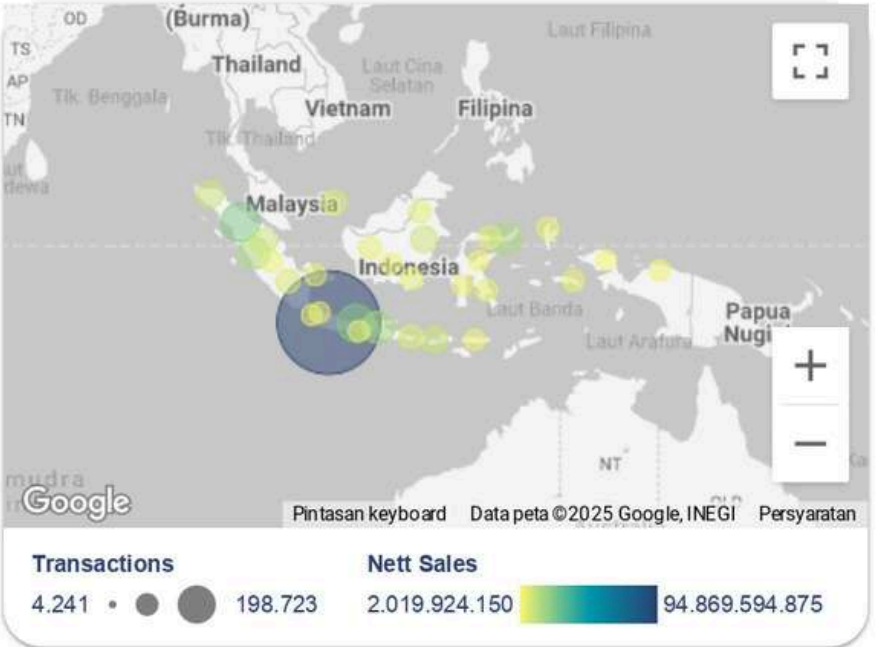
Top Customers Based on Transactions

Customer - ...	Branch	Transact...	Nett Profit	Disc (Avg)	Trans. ...
1. Matthew Smith ...	A - Tomohon	8	Rp957.708,70	6,75%	4,29
2. Michael Smith - ...	A - Sukabumi	6	Rp859.781,90	4,17%	4,18
3. James Smith - ...	A - Purwak...	6	Rp697.097,55	10,00%	4,30
4. Amanda Smith ...	KAL - Cirebon	6	Rp548.493,25	10,67%	3,55
5. John Smith - Ja...	A - Jambi	6	Rp935.393,40	9,33%	3,72

Top 5 High-Rated Branches with Low Transaction Ratings

Branch	Branch ...	Trans....
1. KAL - Pematangsiantar	4,83	3,99
2. KA - Jambi	4,77	4,00
3. KAL - Batam	4,73	4,00
4. A - Sorong	4,73	3,99
5. KAL - Cikampek	4,72	4,02

Transactions & Nett Sales Distribution



Macro-Meso-Micro Framework

The analysis employs a macro-meso-micro framework, starting with broad business questions and progressively narrowing down to specific data challenges. This structure is reflected in the dashboard, organized into three main sections: Overview (Macro), Product Analysis (Meso), and Customer Analysis (Micro). By moving from macro-level insights to detailed meso- and micro-level perspectives, this approach provides a comprehensive understanding of the business and effectively links strategic objectives to actionable insights.

Macro

- How is Kimia Farma performing overall in its business in 2023?
- How did Kimia Farma's revenue trend evolve year over year from 2020 to 2023?
- How is Kimia Farma's profit distributed across different regions in Indonesia?
- How is Kimia Farma's business performance distributed across key provinces?

Meso

- How do transaction volume and revenue move throughout the year?
- How does each branch category contribute to Kimia Farma's overall profit, and which format has the greatest potential for future profit scalability?
- How is Kimia Farma's profit distributed across different product price ranges
- What products and categories primarily drive transaction volume at Kimia Farma?

Micro

- What key factors drive customer loyalty and repeat purchase behavior at Kimia Farma?
- How is the distribution of transactions and net sales across regions?
- What factors drive high branch ratings but relatively low transaction ratings at Kimia Farma?

Macro Analysis – Overview

Q1: How is Kimia Farma performing overall in its business in 2023?

Total Revenue

Rp321,17 M
↑ 33.24%

Total Profit

Rp91,21 M
↑ 30.24%

Avg. Gross Profit Margin

0,26
↑ 0.02%

Kimia Farma's business expansion is reflected in a 33.24% revenue increase to IDR 321.17 billion and a rise in profit to IDR 91.21 billion, with a stable gross profit margin of 0.26, **demonstrating the company's ability to maintain profitability amid growth.**

Total Customer

264,60 rb
↑ 18.69%

Total Transaction

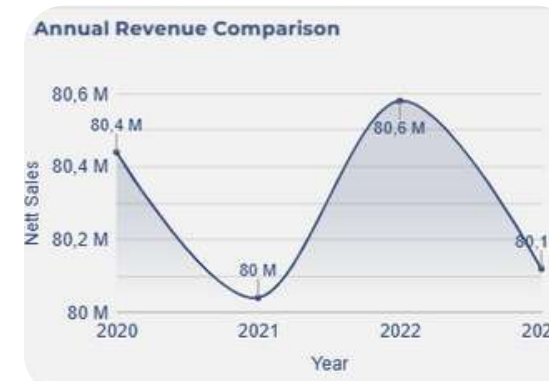
672,46 rb
↑ 33.16%

Avg. Transaction Rating

4,00
↓ -0.01%

An 18.59% increase in the customer base, accompanied by a stronger 33.16% surge in transaction volume, indicates **the company's success in expanding its market reach while also driving higher purchase frequency.** However, a slight 0.01% decline in the average transaction rating to 4.00 signals **the need for continued focus on maintaining and enhancing service quality amid rapid operational expansion.**

Q2: How did Kimia Farma's revenue trend evolve year over year from 2020 to 2023?



Kimia Farma's revenue performance from 2020 to 2023 was **heavily influenced by the COVID-19 pandemic cycle**, with stability in 2020, **a decline in 2021 due to mobility restrictions, supply chain disruptions, and shifts in consumer behavior toward more selective spending on non-essential items**, a strong rebound in 2022 amid economic recovery, and a mild normalization in 2023. This pattern indicates that 2022 growth was largely recovery-driven, while 2023 marks a transition toward a more stable market phase

Recommendation

- Shift from recovery-driven growth to sustainable growth by **strengthening structurally stable segments** such as preventive, chronic care, and wellness products with long-term recurring demand.
- Enhance supply chain resilience through **supplier diversification, improved demand forecasting accuracy, and strategic buffer stock** to safeguard revenue stability during potential disruptions.
- **Accelerate digital and omnichannel expansion** by strengthening e-pharmacy capabilities, loyalty ecosystems, and online-offline (O2O) integration to sustain post-normalization growth.



On a quarterly basis, Kimia Farma's revenue exhibits a **clear seasonality pattern**, strengthening from Q1 to a peak in Q3 before moderating in Q4. **The Q1–Q3 uptrend, which coincides with the rainy-to-dry season transition**, is driven by higher disease incidence, weather shifts, and elevated economic activity that boost demand for healthcare products, **while the Q4 slowdown reflects year-end consumer spending shifts, slower institutional budget realization, and post-peak demand normalization.**

Recommendation

- Front-load revenue acceleration in Q1–Q3 by **prioritizing major promotions, product launches, and healthcare campaigns** during peak demand driven by seasonal diseases and high mobility.
- **Optimize inventory and supply planning** ahead of Q1–Q3 **to prevent stockouts** during demand surges **and avoid excess stock** accumulation entering Q4.
- Shift focus in Q4 from volume growth to margin optimization, by **emphasizing high-margin wellness, chronic care, and private-label products** as demand normalizes.
- Adjust marketing strategy in Q4 toward **retention and loyalty programs**, rather than aggressive acquisition, to stabilize revenue during the natural seasonal slowdown.

Macro Analysis – Overview

Q3: How is Kimia Farma's profit distributed across different regions in Indonesia?



The profit map shows that **Kimia Farma's profit is unevenly distributed across Indonesia and is mainly concentrated in a few core regions**, particularly Java and other areas with high economic activity. The wide profit range from approximately Rp572 thousand to Rp26.94 billion highlights significant regional performance gaps, where **high profit regions are supported by stronger purchasing power, higher population density, and better distribution infrastructure, while low profit regions reflect untapped market potential and access limitations.**

- **Prioritize profit based expansion** by focusing new outlets, capacity upgrades, and marketing investments **on high profit regions to maximize return on investment.**
- **Apply differentiated strategies for low profit regions** by adopting cost efficient models such as **mini outlets, delivery and e pharmacy services, and partnerships with local healthcare facilities.**
- **Implement regional clustering based on profit performance** to define clear strategies for high profit core regions, mid tier efficiency driven regions, and low profit frontier regions.
- Align product mix with regional characteristics by **pushing wellness, chronic care, and private label products in high profit areas** while focusing on **essential and generic medicines in low profit regions.**
- **Optimize geographic inventory allocation by prioritizing fast moving and high margin products to high profit regions** to accelerate cash flow and maximize profit contribution.

Q3: How is Kimia Farma's profit distributed across different regions in Indonesia?



West Java

- ▶ Massive population base, dense urbanization, and intense healthcare activity

North Sumatera & Central Java

- ▶ Form a strong second tier, though their revenue contribution relative to transaction volume suggests lower average basket sizes and a heavier reliance on generic or lower margin products.

East Java

- ▶ Very high transaction volume but comparatively lower revenue, indicating a volume driven market that remains under monetized and presents significant opportunity for up selling and premiumization.

N. Sulawesi, W. Sumatera, Aceh, W. Nusa Tenggara, Riau, E. Kalimantan

- ▶ Contribute relatively smaller shares to total revenue, reflecting a combination of early stage market development and untapped monetization potential despite their underlying economic and industrial strengths (Riau & E. Kalimantan).

Recommendation



West Java

Expand premium outlets by introducing premium health check up and specialized pharmacy services while **accelerating high margin wellness, chronic care, and private label products** through an **integrated omnichannel strategy.**



North Sumatera & Central Java

Increase basket size through **bundled chronic care therapies, subscription medicine programs, and targeted upselling** from generics to branded and wellness products.



East Java

Monetize high transaction volume by **enhancing premiumization, cross selling, and in store health services** to gradually increase revenue per transaction.



N. Sulawesi, W. Sumatera, Aceh, W. Nusa Tenggara

Adopt a low cost, access driven model by **focusing on essential medicines, digital pharmacy, delivery services, and logistics efficiency** rather than heavy physical expansion.



East Kalimantan & Riau

Unlock industrial driven revenue through **B2B healthcare partnerships, corporate pharmacy services, and employee health programs** for the mining, oil, and manufacturing sectors.

How can Kimia Farma reduce its dependency on its core growth region by accelerating revenue growth in second tier and outer regions?

Kimia Farma can **reduce its dependency on West Java** by accelerating potential based expansion in second tier and outer regions through **strengthening last mile distribution networks, expanding e-pharmacy and delivery services to overcome physical access limitations, and establishing strategic partnerships with local hospitals, clinics, and public health centers** to enhance service accessibility, transaction volume, and long term market penetration.

Meso Analysis – Product Analysis

Key Metrics

Highest-Revenue
Product
KF953

Lowest-Revenue
Product
KF172

Highest-Transaction
Product
KF519

Lowest-Transaction
Product
KF451

Highest-Profit Margin
Product
KF519

Lowest-Profit Margin
Product
KF427

KF953 ➤ Drugs for obstructive airway diseases.

KF172 ➤ Psycholeptics drugs, specifically hypnotics and sedative drugs.

KF519 ➤ Analgesics and antipyretics, specifically pyrazolones and anilides.

KF451 ➤ Drugs for obstructive airway diseases.

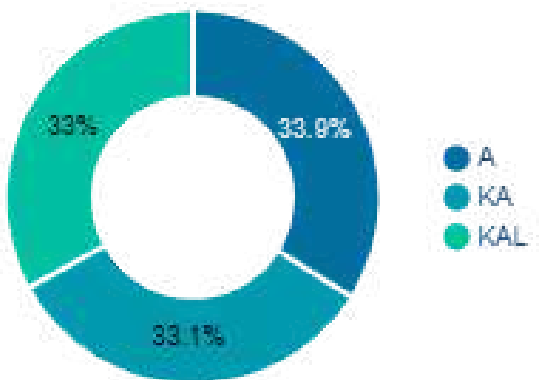
KF427 ➤ Analgesics and antipyretics drugs, salicylic acid and derivatives.

Q1: How do transaction volume and revenue move throughout the year?



- **Revenue in 2023 was primarily volume-driven**, with net sales closely mirroring transaction movement, indicating limited contribution from basket size growth.
- **Monthly seasonality was clear**: sharp post-holiday dip in February, recovery from March–May, mid-year softening, strong rebound in October, and solid performance in December.
- **Future growth should focus on increasing basket size and premiumization** to reduce dependency on transaction volume alone.

Q2: How does each branch category contribute to Kimia Farma's overall profit, and which format has the greatest potential for future profit scalability?



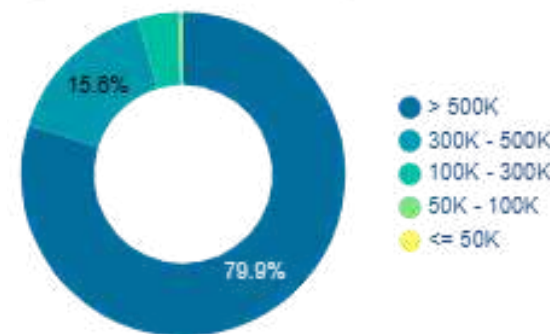
A: Apotek, KA: Klinik-Apotek, KAL: Klinik-Apotek-Lab

Profit contribution across branch categories is evenly distributed, with Apotek at approximately 33.9%, Klinik-Apotek at 33.1%, and Klinik-Apotek-Lab at 33.0%, indicating that no single format currently dominates profitability and that **the business portfolio is well diversified with balanced risk exposure**.

However, this near-equal contribution also suggests that no format has yet been fully optimized as a primary profit accelerator, **reinforcing the need to scale up the most capital-efficient format through ROI-based evaluation and replicate the best-performing model in high-potential regions** to accelerate overall profit growth.

Q3: How is Kimia Farma's profit distributed across different product price ranges

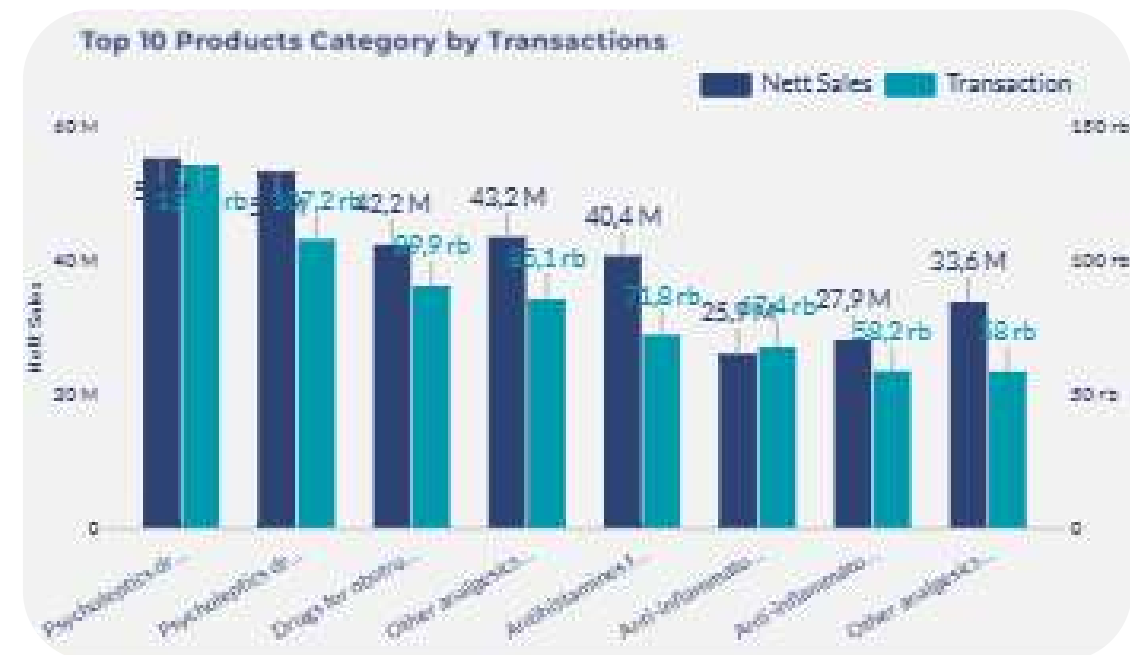
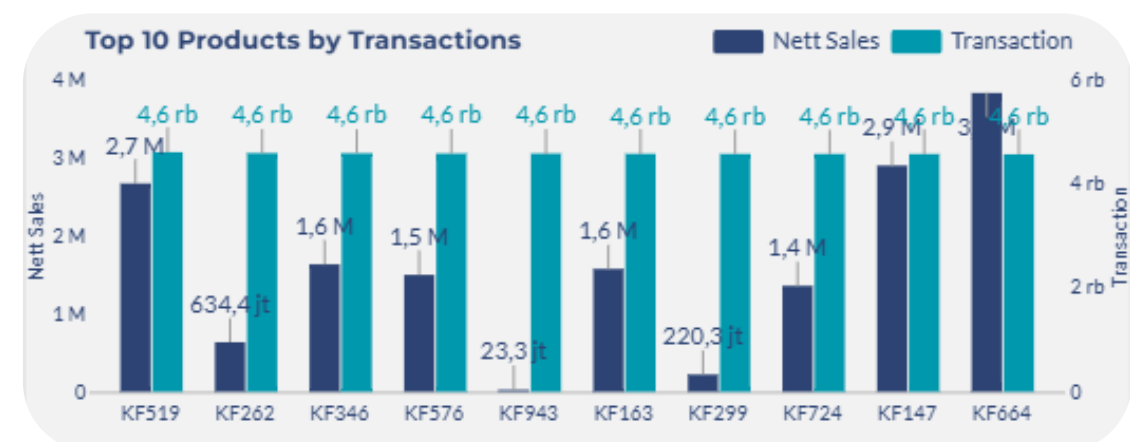
Meso Analysis – Product Analysis



Profit is highly concentrated in the >500K price segment, which contributes approximately 79.9% of total profit, while the 300K–500K segment accounts for only 15.6% and lower price tiers contribute marginally. This structure indicates that **Kimia Farma's profitability is heavily anchored on high value products** such as specialized therapies, medical devices, and premium healthcare services, whereas **lower and mid priced products primarily act as traffic drivers**, revealing a significant monetization gap and increasing exposure to potential purchasing power fluctuations in upper income segments.

- >500k Segment** → Strengthen and scale the premium segment through **strong inventory control, superior service quality, and prioritization of high margin specialized therapies**, medical devices, and premium healthcare services as the main profit driver.
- 300 - 500k Segment** → Upgrade the mid-tier segment through **strategic product bundling, cross-selling from generics to branded and wellness products, and targeted promotions** to structurally migrate demand toward higher-value transactions.
- <300k Segment** → Position the entry-level segment as a **customer acquisition funnel through** high volume penetration, followed by gradual migration to higher value products via education, loyalty programs, and personalized recommendations.

Q4: What products and categories primarily drive transaction volume at Kimia Farma?



- KF519** → Analgesics and antipyretics, pyrazolones and anilides drugs
- KF262** → Psycholeptics, hypnotics and sedatives drugs
- KF346** → Non-steroidal anti-inflammatory drugs (NSAIDs), specifically the acetic acid derivatives subgroup.
- KF576** → Analgesics and antipyretics, salicylic acid drugs

- KF943** → Analgesics and antipyretics, salicylic acid drugs
- KF163** → Non-steroidal anti-inflammatory drugs (NSAIDs), specifically the acetic acid derivatives subgroup.
- KF299** → Psycholeptics, hypnotics and sedatives drugs
- KF724** → Drugs for obstructive airway diseases
- KF147** → Antihistamines for systemic use
- KF664** → Psycholeptics, anxiolytic drugs

Kimia Farma's transaction volume is primarily driven by everyday pain, inflammation, allergy, sleep disorders, anxiety, and airway-related conditions, a pattern that is also reflected in the top product categories dominated by psycholeptics (hypnotics, sedatives, and anxiolytics), respiratory drugs, analgesics–antipyretics, antihistamines, and non-steroidal anti-inflammatory drugs (NSAIDs).

Despite similarly high transaction volumes across the top products, their net sales performance varies significantly, indicating a clear monetization gap within the high-traffic portfolio. **Some products already function as strong revenue generators with high value per transaction, while others primarily act as traffic drivers with relatively low revenue yield, highlighting the need for targeted pricing, bundling, and product positioning strategies to improve overall revenue quality.**

Recommendation

- Prioritize **high-revenue and high-margin products through improved shelf visibility, stronger stock allocation, and active pharmacist recommendations** to maximize return from existing traffic, while ensuring consistent availability of stable-demand categories such as psycholeptics, NSAIDs, and respiratory drugs to sustain traffic.
- Increase monetization of **high-volume, low-yield products through pharmacist-guided recommendations and optimized pack sizes** without compromising patient safety.
- **Strengthen cross-selling strategies** from entry-level products to higher-value chronic care, wellness, and premium formulations to increase basket size **through clinically relevant complementary offerings such as vitamins and wellness supplements.**

Micro Analysis – Customer Analysis

Key Metrics



Customer behavior at Kimia Farma remains volume-driven, as reflected in an **average basket size of 1.0 indicating mostly single-item purchases per visit**. While average basket value (Rp477.61 thousand) and profit per transaction (Rp135.64 thousand) remain stable with slight growth, **the sharp 33.2% increase in customer lifetime value to Rp91.21 million** signals significantly stronger long-term customer value, highlighting clear **upside potential through basket size expansion and cross-selling strategies**.

Q1: What key factors drive customer loyalty and repeat purchase behavior at Kimia Farma?

Customer Segmentation Based on Transactions

	Customer - ...	Tr...	Avg...	Segmentation	Disc (A...	Trans. R...
1.	Michael Smith - ...	13	Rp604.579,38	Loyal Customer	5,54%	3,99
2.	John Smith - B...	12	Rp506.015,00	Loyal Customer	9,75%	3,97
3.	Robert Johnson...	11	Rp472.602,00	Loyal Customer	7,82%	4,37
4.	Michael Smith - ...	11	Rp453.988,27	Loyal Customer	5,55%	4,21
5.	James Smith - ...	11	Rp420.428,55	Loyal Customer	8,18%	4,18

Top Customers Based on Transactions

	Customer - ...	Branch	Transact...	Nett Profit	Disc (Avg)	Trans. ...
1.	Matthew Smith ...	A - Tomohon	8	Rp957.708,70	0,07	4,29
2.	Amanda Smith ...	KAL - Cirebon	6	Rp548.493,25	0,11	3,55
3.	David Williams ...	KAL - Cilacap	6	Rp938.109,60	0,06	4,25
4.	James Smith - ...	A - Purwak...	6	Rp697.097,55	0,10	4,30
5.	Daniel Smith - ...	A - Subang	6	Rp873.648,20	0,07	4,67

The customer segmentation and top customers by transactions show that loyal customers dominate both the highest transaction frequency and profit contribution across multiple branches. **Customers with repeated transactions consistently generate strong net profit even with relatively controlled average discounts, indicating that loyalty is driven more by repeat behavior and perceived value than by aggressive discount dependency.**

The presence of these customers across different branches further highlights the **critical role of store experience, service quality, and local relationship management in sustaining loyalty, rather than pricing incentives alone**. In essence, Kimia Farma's top profit contributors are not one-time buyers, but customers with stable engagement and consistent repeat purchasing patterns.

Recommendation

- **Shift loyalty strategy from discount-driven to experience-driven** by standardizing pharmacist service quality, consultation depth, and speed of service across branches to sustain repeat behavior.
- Strengthen **personalized loyalty programs** based on transaction frequency and purchase history, not only spending value, **to reward true repeat customers and deepen engagement**.

- **Use branch-level repeat purchase and net profit metrics as performance KPIs** to identify high-loyalty branches and replicate their best practices across the network.
- **Apply targeted, data-driven promotions** instead of broad discounts, ensuring incentives support retention without eroding margin.
- Enhance relationship-based selling at the outlet level through **consistent customer recognition, follow-up reminders for chronic care, and proactive pharmacist engagement**.

Q2: How is the distribution of transactions and net sales across regions?



The geographic distribution of transactions and net sales indicates a **clear concentration of customer activity and revenue within core urban regions**, particularly across major provinces in **Java**. **Areas with the largest transaction bubbles do not always correspond proportionally to the highest net sales intensities**, signaling a structural difference between volume-driven regions and value-driven regions.

Micro Analysis – Customer Analysis

This suggests that in **several high-traffic areas, customer behavior is characterized by frequent but lower-value purchases, while other regions, despite lower transaction volume**, contribute disproportionately higher net sales through stronger basket value. This imbalance reflects differences in purchasing power, product mix adoption, and service utilization across regions, implying that **customer value is shaped not only by how often customers buy, but also by what they buy and how they are served**.

Recommendation

- **High-transaction but lower-value regions** should focus on basket size expansion through **clinically guided cross-selling, premium service offerings, and chronic care programs**.
- **High-revenue but lower-transaction regions** should prioritize traffic stimulation through **accessibility, digital ordering, and convenience-driven services**.
- **Regional customer lifetime value and revenue-per-transaction metrics should be embedded as core KPIs** to avoid over-reliance on transaction volume alone.

Q3: What factors drive high branch ratings but relatively low transaction ratings at Kimia Farma?

	Branch	Branch ...	Trans...
1.	KAL - Pematangsiantar	4,83	3,99
2.	KA - Jambi	4,77	4,00
3.	KAL - Batam	4,73	4,00
4.	A - Sorong	4,73	3,99
5.	KAL - Cikampek	4,72	4,02

The consistently **high Branch Ratings (around 4.75) reflect strong customer perceptions of the physical environment and human interaction**, including branch cleanliness, location, staff friendliness, product knowledge, and overall atmosphere. In contrast, **the lower Transaction Ratings (around 4.00) signal friction within the service delivery process itself, such as prolonged waiting times, workflow inefficiencies, digital or self-service limitations, and procedural or system-related bottlenecks**. This gap highlights a clear misalignment between strong service experience and operational execution, which constrains the branch’s ability to convert service excellence into stronger transactional performance.

Recommendation

- Streamline transaction workflows by **reducing unnecessary steps, simplifying procedures, and standardizing service** processes to shorten waiting times.
- **Strengthen digital and self-service channel reliability** to minimize system delays, input errors, and transaction bottlenecks.
- **Implement real-time queue monitoring and staff reallocation** during peak hours to balance service load and improve transaction speed.
- **Equip frontliners and pharmacists with soft-selling and service-to-conversion training** to translate strong service perception into higher transaction value without compromising trust.
- **Track transaction efficiency KPIs at branch level (waiting time, transaction completion time, transaction per visit)** alongside service ratings to ensure service excellence is supported by operational performance.



Thank You!



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